Exercise 2.1

SOLUTION

نوٹ
دیے گئے سوال میں X کی جگہ قیمت لگانے سے جواب آجاتا ہے
مثلاً اگر $f(x)=x^2+1$ تو $f(0)$ معلوم کرنے کے لیے x کی جگہ 0 لگائیں

Q1: Given that

(a) $f(x)=x^2-1$ (b) $f(x)=\sqrt{2x+3}$

(i) To find $f(-3)$

Sol: Put $x=-3$ into the function

$\Rightarrow f(-3)=(-3)^2-1=9-1=8$

(ii) To find $f(0)$

Sol: Put $x=0$ into the function

$\Rightarrow f(0)=(0)^2-1=0-1=-1$

(iii) To find $f(x-2)$

Sol: Replacing x by $x-2$ into the function

$\Rightarrow f(x-2)=(x-2)^2-1$

$=x^2-4x+4-1=x^2-4x+3$

(iv) To find $f(x^2+3)$

Sol: Replace x by x^2+3 into the function

$\Rightarrow f(x^2+3)=(x^2+3)^2-1$

$=x^4+6x^2+9-1=x^4+6x^2+8$

(b) Given that $f(x)=\sqrt{2x+3}$

(i) To find $f(-3)$

Sol: Put $x=-3$ into the function

$\Rightarrow f(-3)=\sqrt{2(-3)+3}=\sqrt{-6+3}=\sqrt{-3}$

(ii) To find $f(0)$

Sol: Put $x=0$ into the function

$\Rightarrow f(0)=\sqrt{2(0)+3}=\sqrt{0+3}=\sqrt{3}$

(iii) To find $f(x-2)$

Sol: Put replacing $f(x-2)$ into the function

$\Rightarrow f(x-2)=\sqrt{2(x-2)+3}$

$=\sqrt{2x-4+3}=\sqrt{2x-1}$

(iv) To find $f(x^2+3)$

Sol: Put $f(x^2+3)$ into the function

$\Rightarrow f(x^2+3)=\sqrt{2(x^2+3)+3}$

$=\sqrt{2x^2+6+3}=\sqrt{2x^2+9}$

Q2: Find $\dfrac{f(a+h)-f(a)}{h}$ and simplify where,

(i) $f(x)=4x+7$

Sol: Here we have to find $f(a)$ and $f(a+h)$.

As;

$f(a)=4a+7$ and $f(a+h)=4(a+h)+7.$

$f(a+h)=4a+4h+7.$

$\dfrac{f(a+h)-f(a)}{h}=\dfrac{4a+4h+7-4a-7}{h}=\dfrac{4h}{h}=4$

(ii) $f(x)=\sin x,$

Sol: As $f(x)=\sin x$

Then $f(a)=\sin a$ and $f(a+h)=\sin(a+h)$

Now, $\dfrac{f(a+h)-f(a)}{h}=\dfrac{\sin(a+h)-\sin a}{h}$

$\therefore \sin P-\sin Q=2\cos\dfrac{P+Q}{2}\cdot\sin\dfrac{P+Q}{2}$

$=\dfrac{2\cos\dfrac{a+h+a}{2}\cdot\sin\dfrac{a+h-a}{2}}{h}$

$=\dfrac{2\cos\left(\dfrac{2a+h}{2}\right)\cdot\sin\left(\dfrac{h}{2}\right)}{h}$

$=\dfrac{2}{h}\cos\left(a+\dfrac{h}{2}\right)\cdot\sin\left(\dfrac{h}{2}\right)$

(iii) $f(x)=x^3+x^2-1$, Finding $f(a)$ and $f(a+h)$

Sol: As; $f(a)=a^3+a^2-1$

$f(a+h)=(a+h)^3+(a+h)^2-1$

$=a^3+h^3+3a^2h+3ah^2+a^2+2ah+h^2-1$

Now, $\dfrac{f(a+h)-f(a)}{h}$

$=\dfrac{a^3+h^3+3a^2h+3ah^2+a^2+2ah+h^2-1-a^3-a^2+1}{h}$

$=\dfrac{h\left(h^2+3a^2+3ah+2a+h\right)}{h}$

$=h^2+3ah+h+3a^2+2a$

CamScanner

(iv) $f(x)=\tan x$, Calculating $f(a)$ and $f(a+h)$.

Sol: As $f(a)=\tan a$ and $f(a+h)=\tan(a+h)$

Now, $\dfrac{f(a+h)-f(a)}{h}=\dfrac{\tan(a+h)-\tan a}{h}$

$$=\frac{1}{h}\left(\frac{\sin(a+h)}{\cos(a+h)}-\frac{\sin a}{\cos a}\right)$$

$$=\frac{1}{h}\left(\frac{\sin(a+h)\cos a-\cos(a+h)\sin a}{\cos(a+h)\cos a}\right)$$

$\therefore \sin\alpha\cos\beta-\cos\alpha\sin\beta=\sin(\alpha-\beta)$

$$=\frac{1}{h}\left(\frac{\sin(a+h-a)}{\cos(a+h)\cos a}\right)$$

$$=\frac{1}{h}\left(\frac{\sin h}{\cos(a+h)\cos a}\right)=\frac{\sin h}{h\cos(a+h)\cos a}$$

Q3: Express the following:

a) The area A of a square as a function of its perimeter P.

b) The circumference C of a circle as a function of its area A.

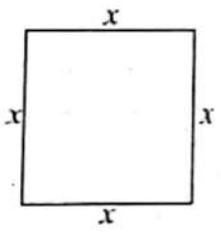
c) The surface area S of a cube as a function of its volume V.

(a) Let x be the side of square.

Sol: Then area A will be: $A=x^2 \longrightarrow$ (i)

The perimeter (boundary wall distance) of square is given as:

$$P=4x \longrightarrow \quad \text{(ii)}$$



From equation (ii)

$$x=\frac{P}{4} \longrightarrow \quad \text{(iii)}$$

Putting in equation (i)

$$A=\left(\frac{P}{4}\right)^2 \Rightarrow A=\frac{P^2}{16}$$

(b) Let r be the radius of circle.

Sol: The circumference C (Perimeter of circle) is given as

$C=2\pi r \longrightarrow$ (i)

The area of circle is given as:

$$A=\pi r^2 \longrightarrow \text{(ii)}$$

From equation (ii)

$$r^2=\frac{A}{\pi} \Rightarrow r=\sqrt{\frac{A}{\pi}} \longrightarrow \quad \text{(iii)}$$

Putting in equation (iii) in equation (i)

$$C=2\pi\sqrt{\frac{A}{\pi}}=2\sqrt{\pi A}$$

(c) Let the length of the side of the cube is l.

Sol: Surface area of the cube is given as.

$$S=6(l\times l)=6l^2 \longrightarrow \quad \text{(i)}$$

The volume of the cube is given as:

$$V=l^3 \longrightarrow \quad \text{(ii)}$$

> **Note:**
> Cube is a diagram whose length, width, height are equal.

From equation (ii)

$$l=(V)^{\frac{1}{3}} \longrightarrow \quad \text{(iii)}$$

Putting in equation (i)

$$S=6\left(V^{\frac{1}{3}}\right)^2=6V^{\frac{2}{3}}=6\sqrt[3]{V^2}$$

Q4: Find the domain and the range of the function g defined below:

(i) $g(x)=5-x$

Sol: The function is real valued so, its domain and range are:

Domain of $g(x)=\mathbb{R}=(-\infty,\infty)$ = (Set of all real numbers)

Range of $g(x)=\mathbb{R}=(-\infty,\infty)$ = (Set of all real numbers)

(ii) $g(x)=\sqrt{x+2}$

Sol: The given function is radical function in which radicand must be positive.

For domain, using the condition as: $x+2\geq 0$

> نوٹ
> اگر دیے گئے سوال یا فنکشن میں x کے ساتھ تقسیم نہیں ہو رہا
> اور جذر بھی نہیں ہے تو اس کی Domain
> Set of Real number ہے۔

$\Rightarrow \quad x\geq -2 \quad$ or $\quad x\in[-2,\infty)$

Domain of $g(x)=[-2,\infty)$

For range, $\sqrt{x+2}\geq 0$ as $x+2\geq 0$

Range of $g(x)=[0,\infty)$

(iii) $g(x)=\begin{cases}6x+7, & x\leq -2\\ 4-3x, & x>-2\end{cases}$

Sol: It is a piecewise function and working for two conditions.

When input is $x\leq -2 \quad$ or $\quad x\in(\infty,-2]$

Then $g(x)=6x+7$ is working in this case output of the functions are:

$g(-2)=6(-2)+7=-12+7=-5$

CamScanner

$g(-3)=6(-3)+7=-18+7=-11$

Order in output values on real line is less than so, the range for this piece is $(-\infty,-5]$.

When input is $x>-2$ or $x\in(-2,\infty)$

Then $g(x)=4-3x$ is working in this case output of the functions are: $g(-2)=4-3(-2)=4+6=10$

$g(-1)=4-3(-1)=4+3=7$

Order in output values on real line is less than so, the range for this piece is $(-\infty,10)$.

Domain of $g(x)=(-\infty,-2]\cup(-2,\infty)=(-\infty,\infty)=\mathbb{R}$

Range of $g(x)=(-\infty,-5]\cup(-\infty,10)$

Range of $g(x)=(-\infty,10)$

> نوٹ
>
> اگر x ساتھ تقسیم ہو رہا ہے تو x کی جس قیمت یا Denominator
>
> 0 بنے گا اس کو $\mathbb{R}$ میں سے منفی کرنے سے Domain آجائے گی۔

(iv) $g(x)=|x-5|$,

Sol: It is an absolute value function (Whose Range must be the positive real number)

For domain $|x-5|\geq 0 \Rightarrow \pm(x-5)\geq 0$

$x-5\geq 0 \Rightarrow x\geq 5$

$x-5\leq 0 \Rightarrow x\leq 5$

Domain of $g(x)=(-\infty,5]\cup(5,\infty)=(-\infty,\infty)=\mathbb{R}$

Range of $g(x)=[0,\infty)$ = set of all non-negative real numbers

(v) $g(x)=\dfrac{x+2}{3-x}$

Sol: $g(x)=\dfrac{x+2}{3-x}$

Domain of $g(x)=\mathbb{R}-\{3\}=(-\infty,3)\cup(3,\infty)$

Range of $g(x)=\mathbb{R}-\{-1\}=(-\infty,-1)\cup(-1,\infty)$

Q5: **Given $f(x)=x^3-ax^2+bx+1$. If $f(2)=-3$ and $f(-1)=0$ Find the values of a & b**

Sol: Given function is $f(x)=x^3-ax^2+bx+1\longrightarrow(i)$

Given $f(2)=-3$

$\Rightarrow f(2)=(2)^3-a(2)^2+b(2)+1=-3$

$8-4a+2b+1=-3$

$9-4a+2b=-3$

$\Rightarrow 4a-2b=12$ or $2a-b=6\longrightarrow(ii)$

Also given $f(-1)=0$

$\Rightarrow f(-1)=(-1)^3-a(-1)^2+b(-1)+1=0$

$-1-a-b+1=0$

$-a-b=0\Rightarrow a+b=0$

$\Rightarrow a=-b\longrightarrow(iii)$

Putting equation (iii) into equation (ii)

$2(-b)-b=6 \Rightarrow -2b-b=6$

$-3b=6 \Rightarrow b=-2$

From equation (iii)

$a=-(-2) \Rightarrow a=2$

Q6: **A stone falls from a height of 60m on the ground, the height h after x seconds is approximately given by $h(x)=40-10x^2$.**

i. **What is the height of the stone when:**

a. $x=1\sec$? **b.** $x=1.5\sec$? **c.** $x=1.7\sec$?

---

ii. **When does stone strike the ground?**

Sol: The height h after x sec. is given as:

$h(x)=40-10x^2\longrightarrow$ (i)

$\therefore$ (Height is the function of time)

(i) **a.** When $x=1\sec$

Then $h(1)=40-10(1)^2=40-10=30m$

b. When $x=1.5\sec$

Then $h(1.5)=40-10(1.5)^2=40-10(2.25)$

$=40-22.5=17.5m$

c. When $x=1.7\sec$

Then $h(1.7)=40-10(1.7)^2=40-10(2.89)$

$=40-28.9=11.1m$

(ii) **When stone strikes the ground its height is zero.**

Then $h(x)=0\Rightarrow 40-10x^2=0\Rightarrow 10x^2=40$

$\Rightarrow x^2=4 \Rightarrow x=2\sec$

Q7: **Consider the function $f(x)=3x-5$.**

i. **Determine the domain and range of $f(x)$.**

ii. **Is the function f one-to-one? Justify your answer.**

iii. **Is the function f onto if the co-domain is all real numbers? Explain.**

Sol: Given function $f(x)=3x-5\longrightarrow$ (i)

(i) For domain it is clear that any of the real number can be put into their function

Domain of $f(x)=\mathbb{R}=(-\infty,\infty)$

Range of $f(x)=\mathbb{R}=(-\infty,\infty)$

CamScanner

(ii) The function $f(x)=3x-5 \longrightarrow$ (i)

Will be one-to-one if there is unique output for each and every input.

When $f(x_1)=f(x_2) \Rightarrow x_1=x_2$

$\Rightarrow 3x_1-5=3x_2-5 \Rightarrow x_1=x_2$

So $f(x)$ is one-to-one function for $x \in \mathbb{R}$

(iii) **Given that the co-domain is the set of all real numbers**

R = set of real numbers

R = co-domain of f

So the function $f(x)$ is onto

Q8: **Left $f: R \to R$ be defined by $f(x)=\dfrac{2x-3}{x+1}$**

i. **Find the domain and range of $f(x)$.**

ii. **Determine whether $f(x)$ is onto.**

iii. **Prove that $f(x)$ is one-to-one.**

Sol (i): For Domain

As $f(x)=\dfrac{2x-3}{x+1} \longrightarrow$ (i) be the rational function defined by $f: \mathbb{R} \to \mathbb{R}$.

Which shows $f(x)$ is real valued function.

$f(x)=\dfrac{2x-3}{x+1}$ can exist if $x \neq -1$.

Domain of $f(x)=\mathbb{R}-\{-1\}=(-\infty,-1)\cup(-1,\infty)$

(all real number except -1)

For Range

Let

$$y=\frac{2x-3}{x+1}$$

$$xy+y=2x-3$$

$$xy-2x=-y-3$$

$$x(y-2)=-y-3$$

$$x=\frac{-y-3}{y-2}$$

Range of $f=\mathbb{R}-\{2\}$

(ii) Let

$$y=\frac{2x-3}{x+1}$$

$$y(x+1)=2x-3$$

$$yx+y=2x-3$$

$$yx-2x=-y-3$$

$$x(y-2)=-(y+3)$$

If $y=2$ then $0=-5$

Which is Impossible

Hence $y=2$ is not in the range of $f(x)$

Therefore

$f(x)$ is not onto

(iii) The given function $f(x)=\dfrac{2x-3}{x+1} \longrightarrow$ (i) is called one-to-one if it assigns a unique value for each and every input.

When $f(x_1)=f(x_2) \Rightarrow x_1=x_2 \longrightarrow$ (ii)

So, $f(x_1)=\dfrac{2x_1-3}{x_1+1}$, $f(x_2)=\dfrac{2x_2-3}{x_2+1}$

Using the condition $f(x_1)=f(x_2)$

$$\frac{2x_1-3}{x_1+1}=\frac{2x_2-3}{x_2+1}$$

$$\Rightarrow (2x_1-3)(x_2+1)=(x_1+1)(2x_2-3)$$

$$\cancel{2x_1x_2}+2x_1-3x_2-\cancel{3}=\cancel{2x_1x_2}-3x_1+2x_2-\cancel{3}$$

$$2x_1+3x_1=2x_2+3x_2$$

$$5x_1=5x_2$$

$$\Rightarrow x_1=x_2$$

Prove that $f(x)$ is one-to-one function

Q9: **Consider the function $f: \mathbb{R}^+ \to \mathbb{R}^+$ defined by $f(x)=e^{-x}$. Show that $f(x)$ is a bijective**

Sol: This function will be bijective if it is one-to-one and onto.

When $f(x_1)=f(x_2) \Rightarrow x_1=x_2$

Then $e^{-x_1}=e^{-x_2}$

$-x_1=-x_2 \Rightarrow x_1=x_2$

This shows that $f(x)=e^{-x} \to$ (i) is one-to-one

Now for onto

$f(x)=y \to$ (ii) when $y=e^{-x} \to$ (iii)

From (iii) $\ln y=-x$ or $x=-\ln y$

$f(x)=f(-\ln y)=e^{-(\ln y)}=e^{\ln y}=y$

$f(x)=y$, This shows that $f(x)=e^{-x}$ is onto function.

So, $f(x)=e^{-x}$ is bijective functions

CamScanner

Q10: Let $g : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^3 - 3x$. Determine if $g(x)$ is injective and/ or surjective.

Sol: $f(x_1) = f(x_2)$ where $x_1 = x_2$

$\Rightarrow \quad x_1^3 - 3x_1 = x_2^3 - 3x_2$

$x_1^3 - x_2^3 = 3x_1 - 3x_2$

$(x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 3(x_1 - x_2)$

$x_1^2 + x_1x_2 + x_2^2 = 3 \Rightarrow \quad x_1 \neq x_2$

For example, $f(1) = (1)^3 - 3(1) = -2$

When $x_1 = 1$, then $y = f(1) = -2$

When $x_2 = -2$, then $y = f(-2) = -2$

As $x_1 \neq x_2$, while $f(x_1) = f(x_2)$

$f(x)$ is not injective function.

When $y = g(x)$

Then $y = x^3 - 3x$

As x cannot be expresses in term of y so $g(x)$ cannot be onto function $\Rightarrow$ ($f^{-1}(y)$ does not exist). So $g(x)$ is surjective.

CamScanner